Ejercicio 15. La empresa Loquevenga S.A. ha decidido implementar una red de comunicación de datos, para facilitar el trabajo de su personal, pero, preocupada por los riesgos de que los trabajadores del área de personal pueda ver información sensible del área de contabilidad, diseña una red con la topología que se muestra en el siguiente gráfico, en la que se definen 2 VLANs, cada una de las cuales se organiza de la siguiente manera: VLAN contabilidad: Pc01 Pc02 Pc03; y VLAN personal: Pc10 Pc11 Pc12. La conexión de los equipos a los switches se muestra en el diagrama. Indique la configuración necesaria en los switches de manera que los equipos funcionen de acuerdo a los requerimientos establecidos.

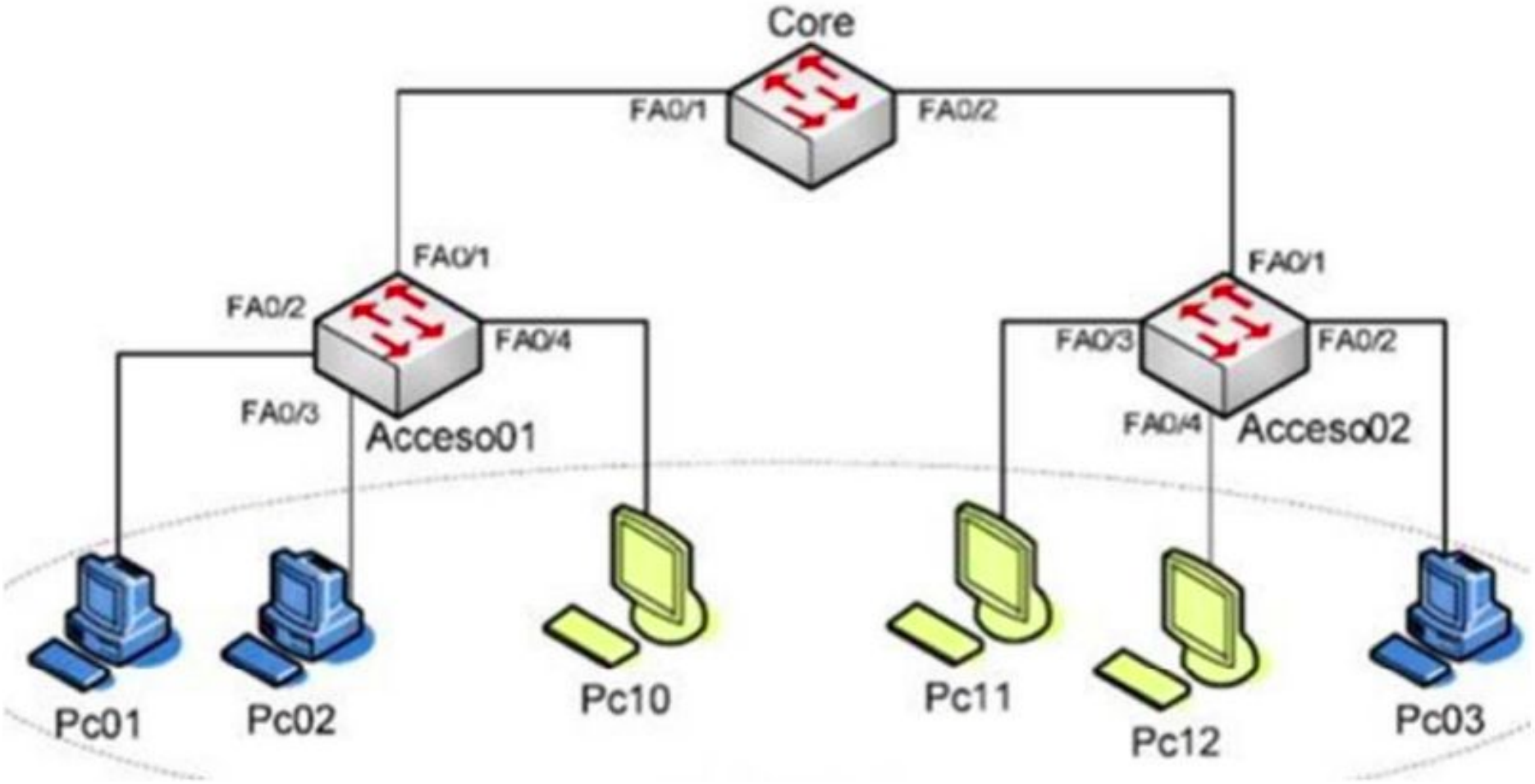
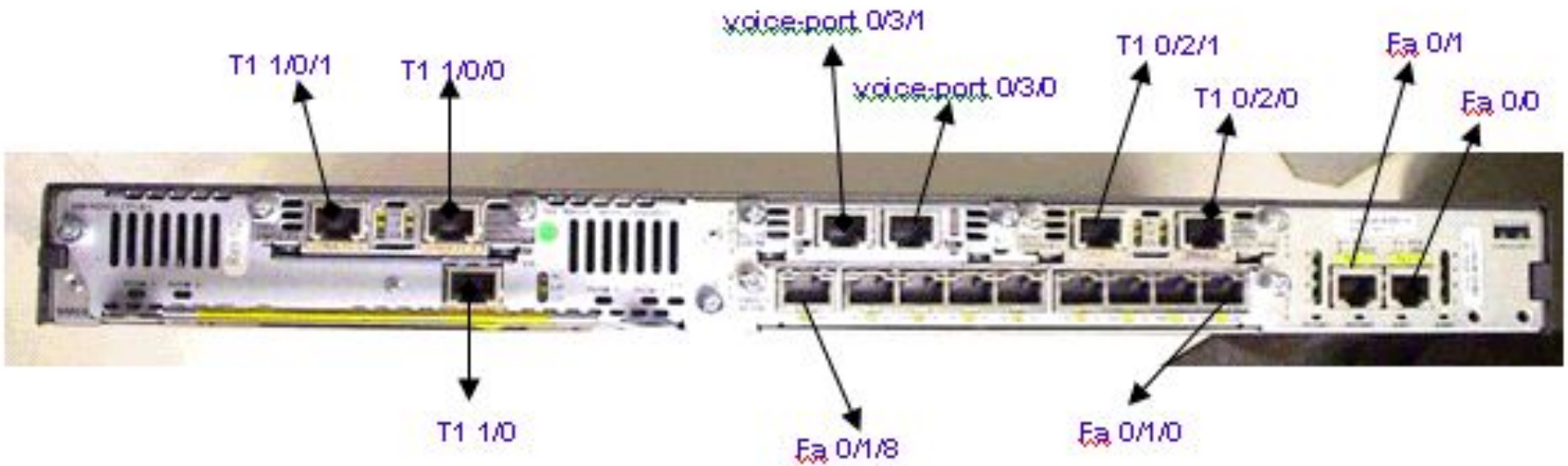
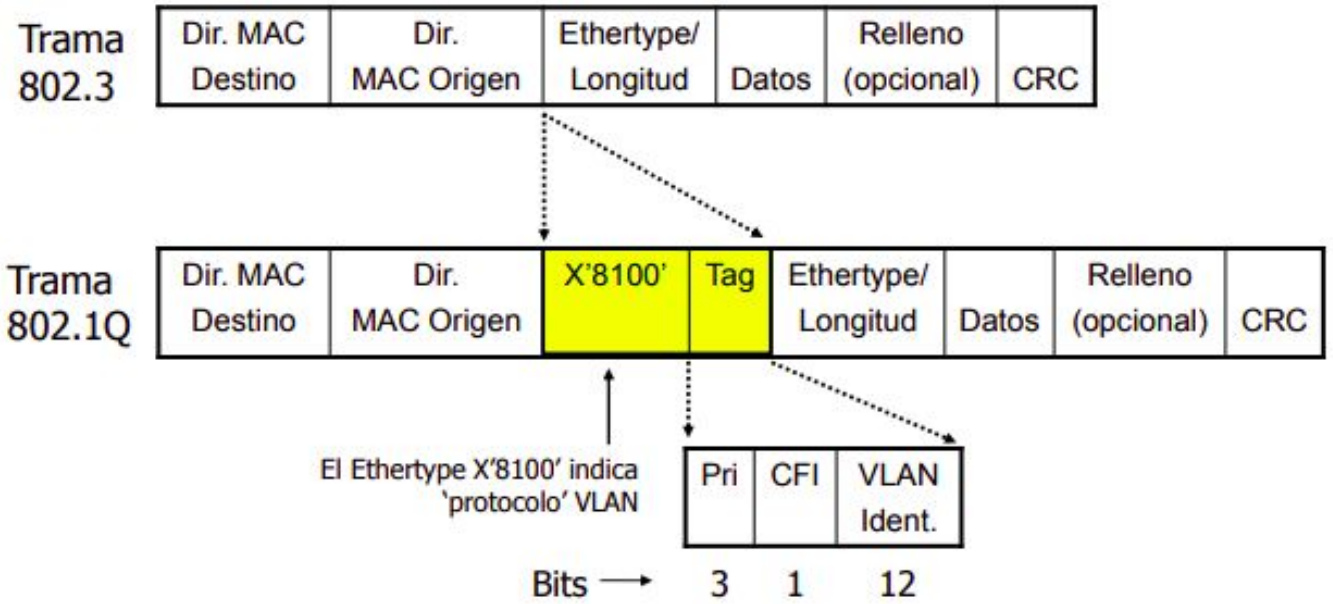

Core
FA0/1
FA0/2
FA0/1
FA0/2
FA0/4
FA0/3
Acceso01
FA0/1
FA0/3
FA0/2
FA0/4
Acceso02
Pc01
Pc02
Pc10
Pc11
Pc12
Pc03

Figure 1 – Interface Numbering on Cisco 2811 Platform

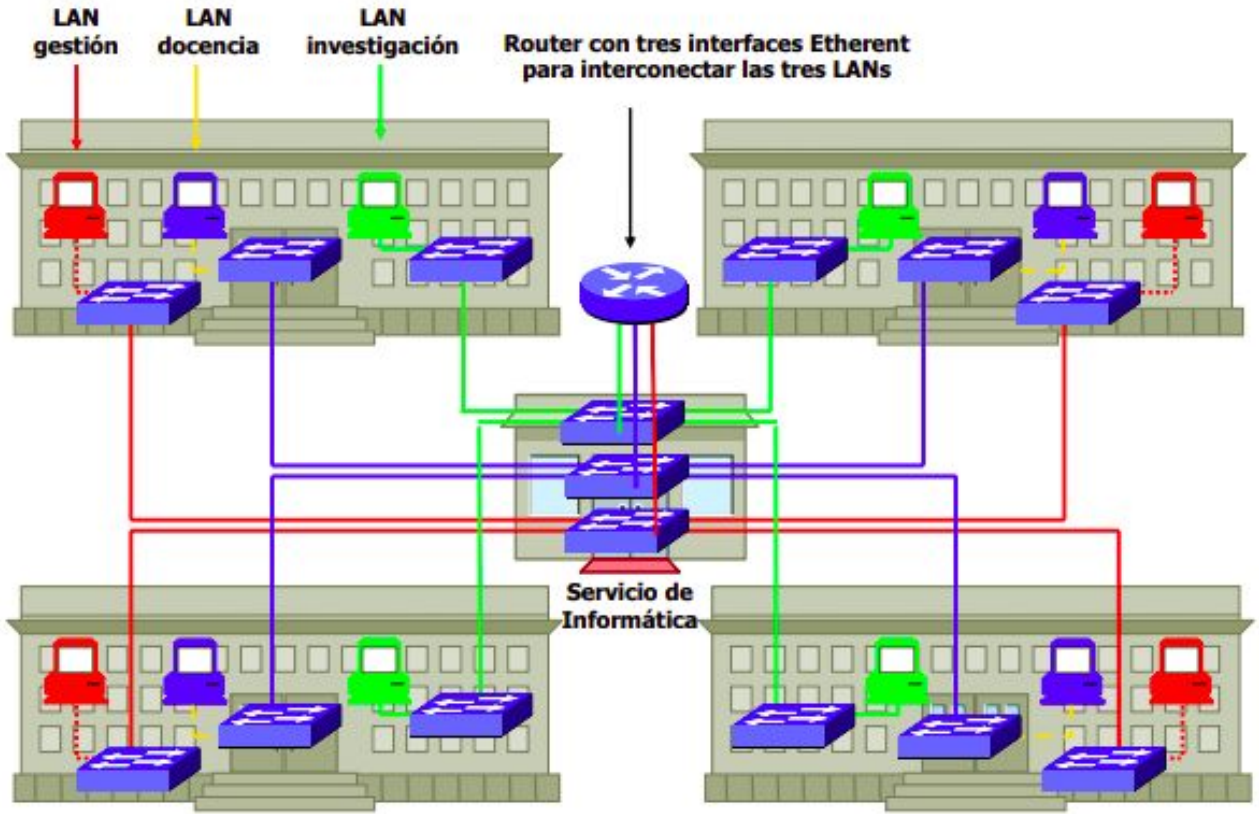


IEEE 802.1Q

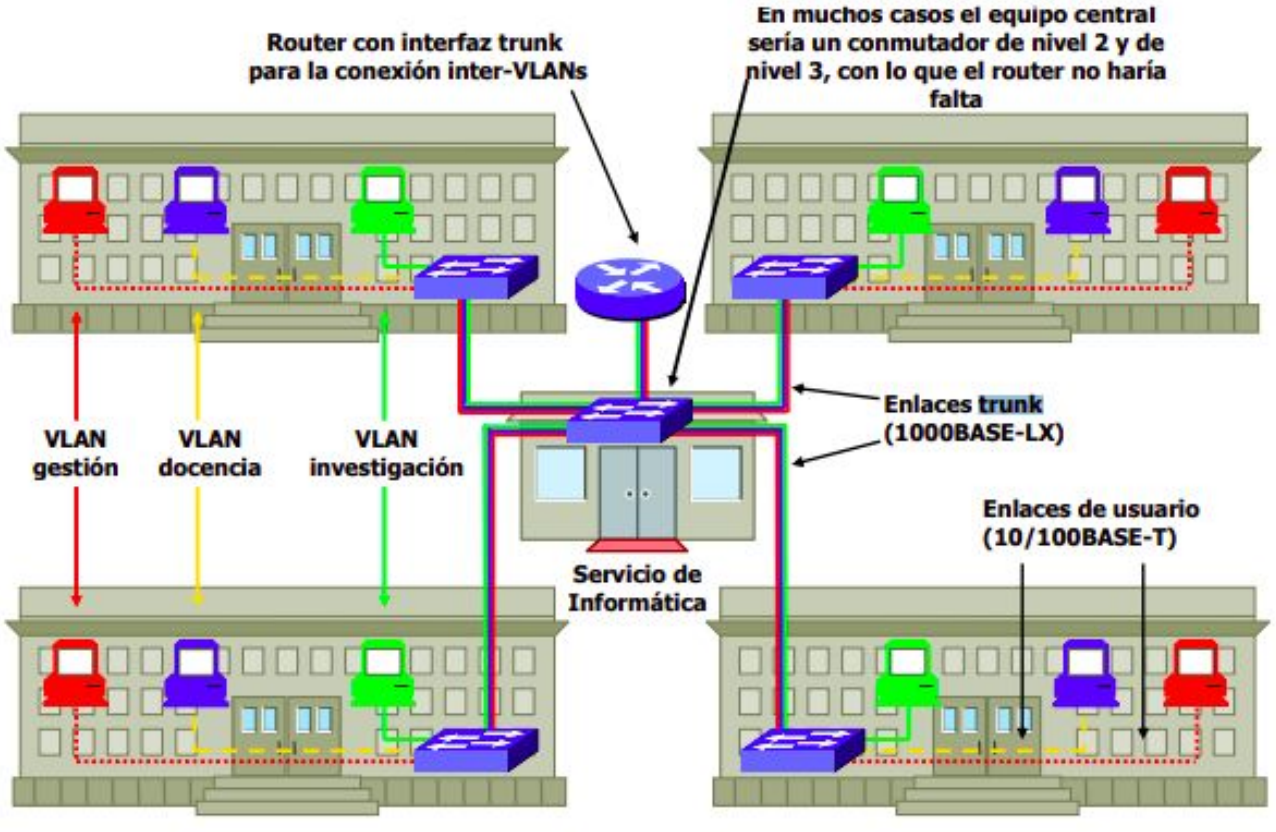


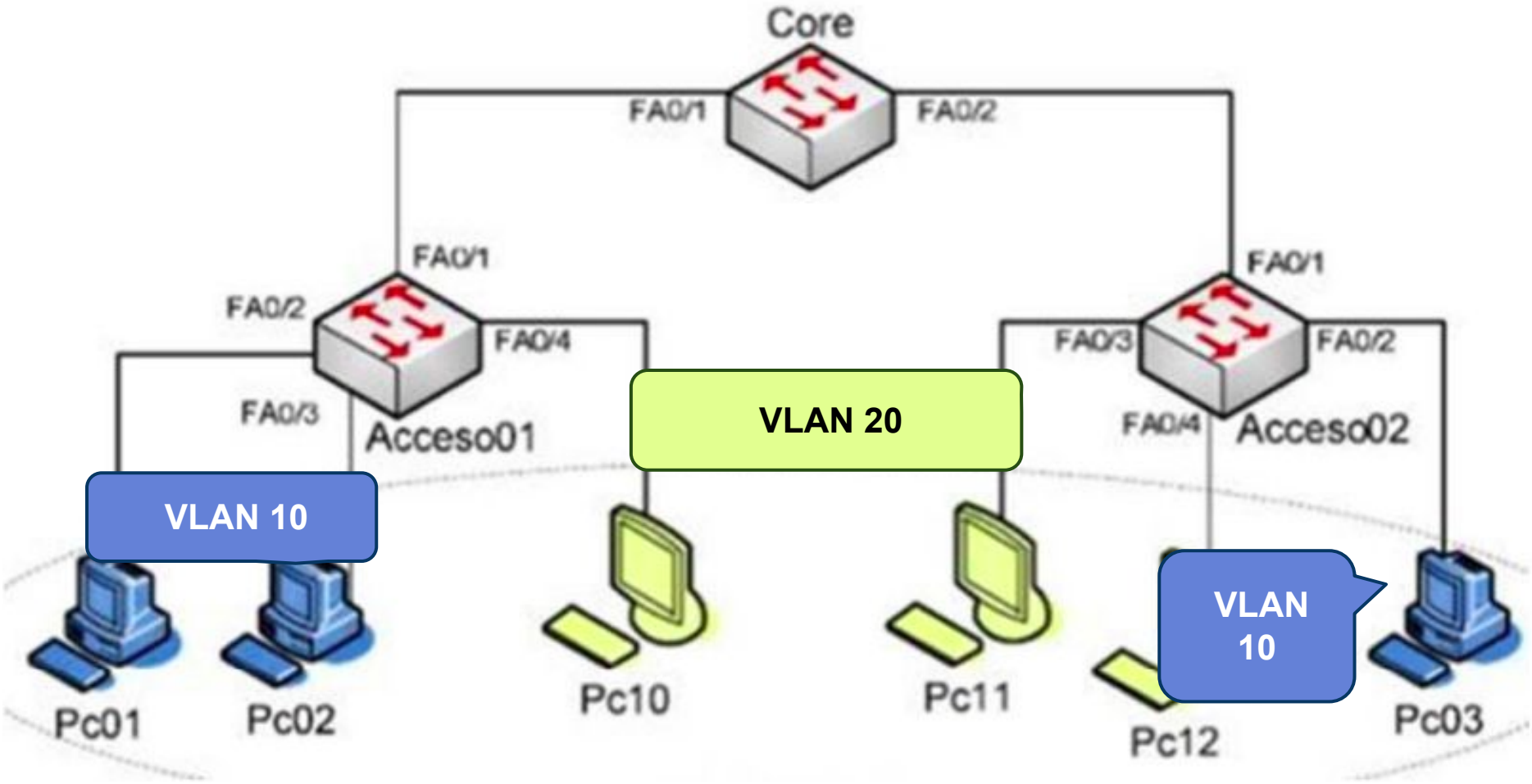
Pri: Prioridad (8 niveles posibles)
CFI: Canonical Format Indicator (indica formato de direcciones MAC)
VLAN Ident.: Identificador VLAN (máximo 4096 en una misma red)

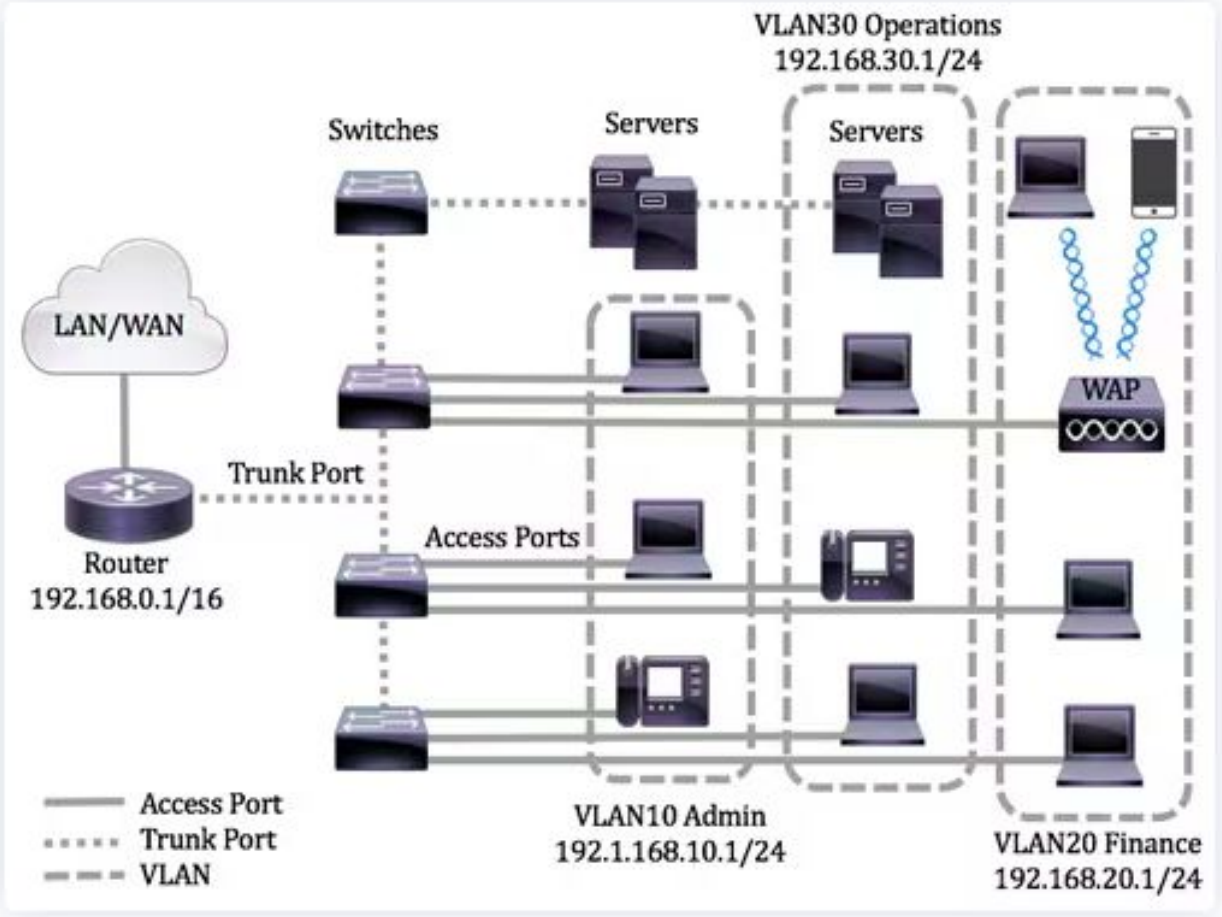
Red con tres LANs



Red con tres VLANs



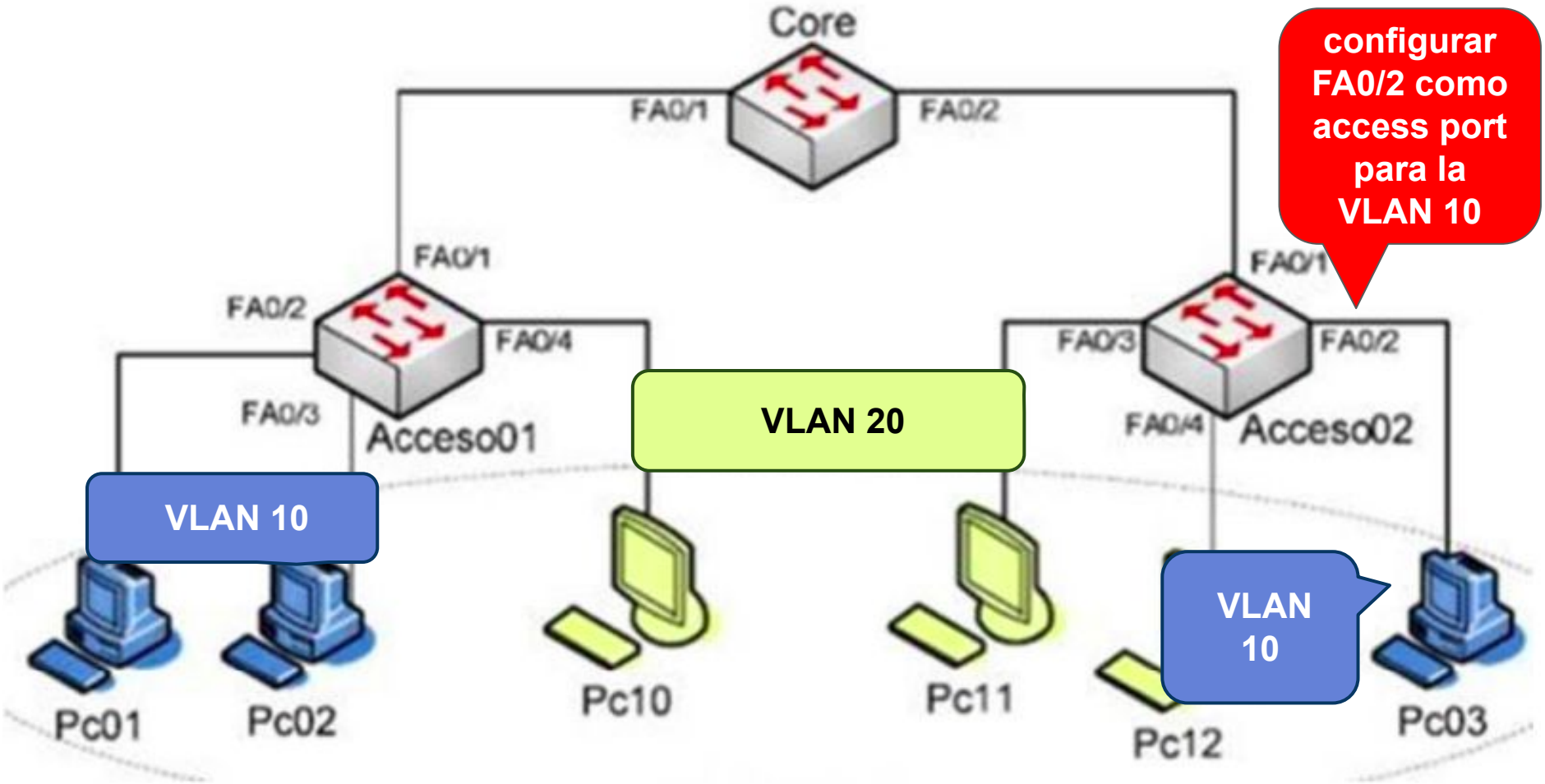

Core
FA0/1
FA0/2
FA0/1
FA0/2
FA0/4
FA0/3
Acceso01
VLAN 20
FA0/1
FA0/3
FA0/2
FA0/4
Acceso02
VLAN 10
VLAN 10
Pc01
Pc02
Pc10
Pc11
Pc12
Pc03

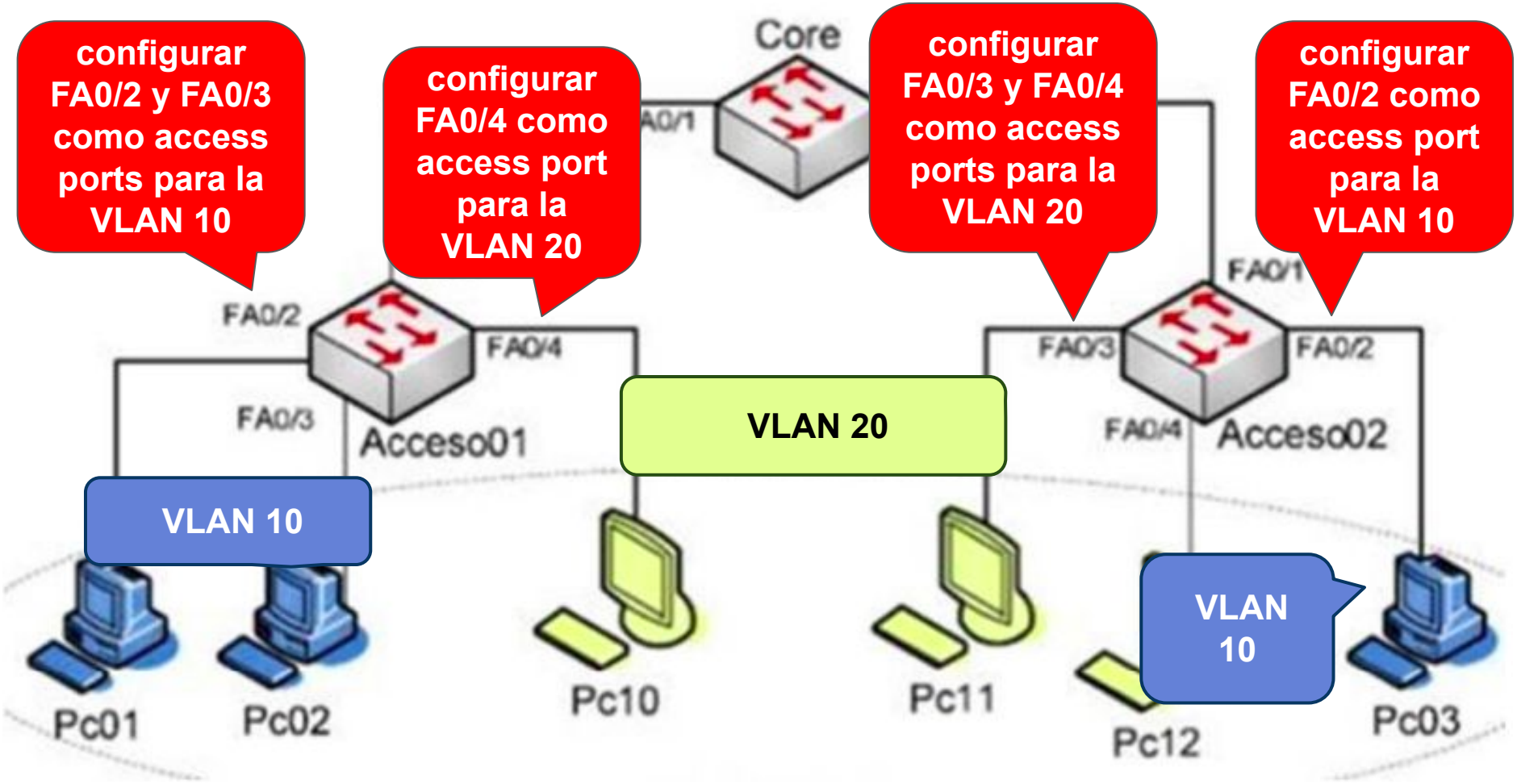

VLAN30 Operations
192.168.30.1/24
Switches
Servers
Servers
LAN/WAN
WAP
Trunk Port
Access Ports
Router
192.168.0.1/16
Access Port
Trunk Port
VLAN
VLAN10 Admin
192.1.168.10.1/24
VLAN20 Finance
192.168.20.1/24

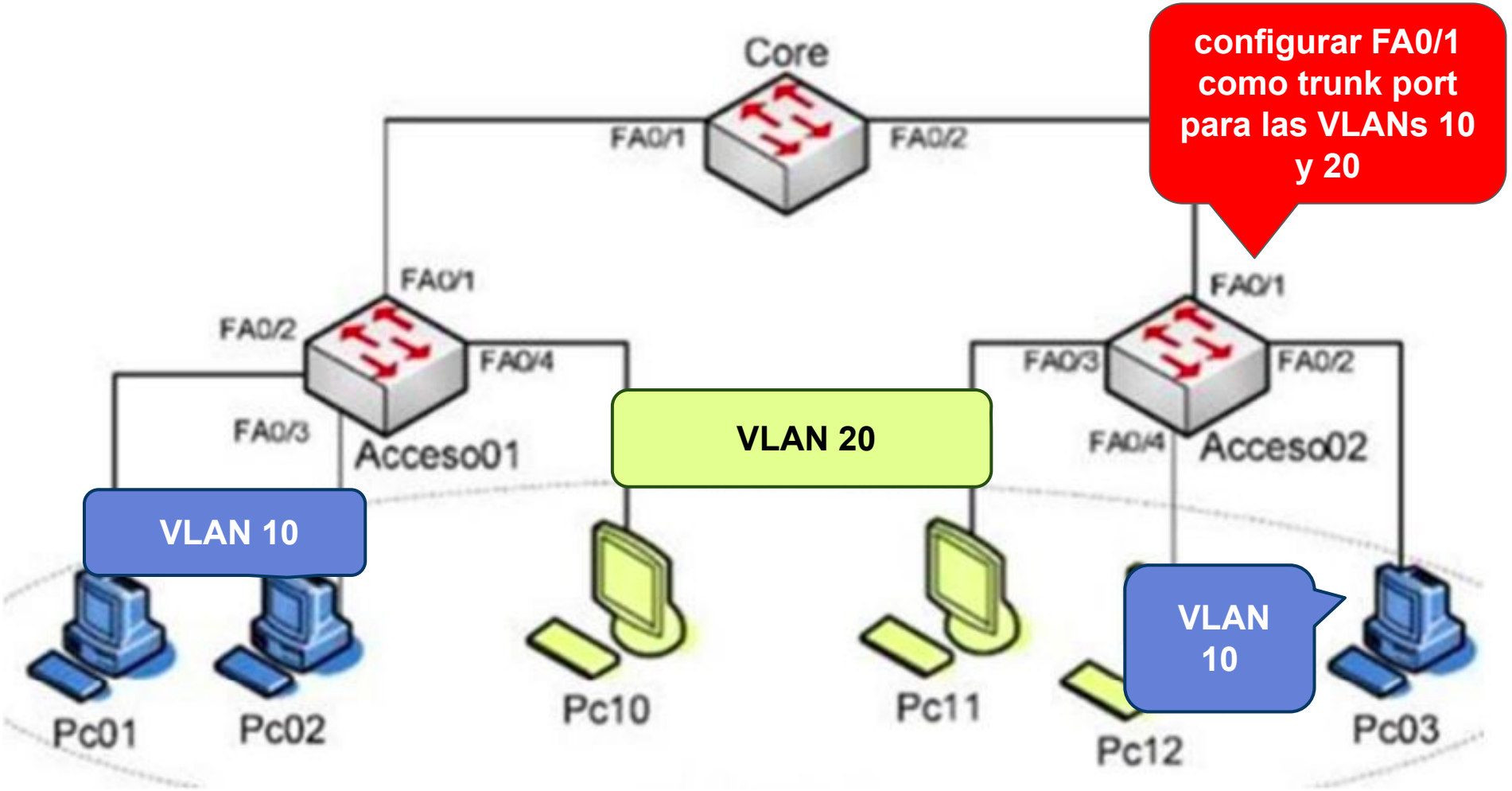
The port modes are defined as follows:

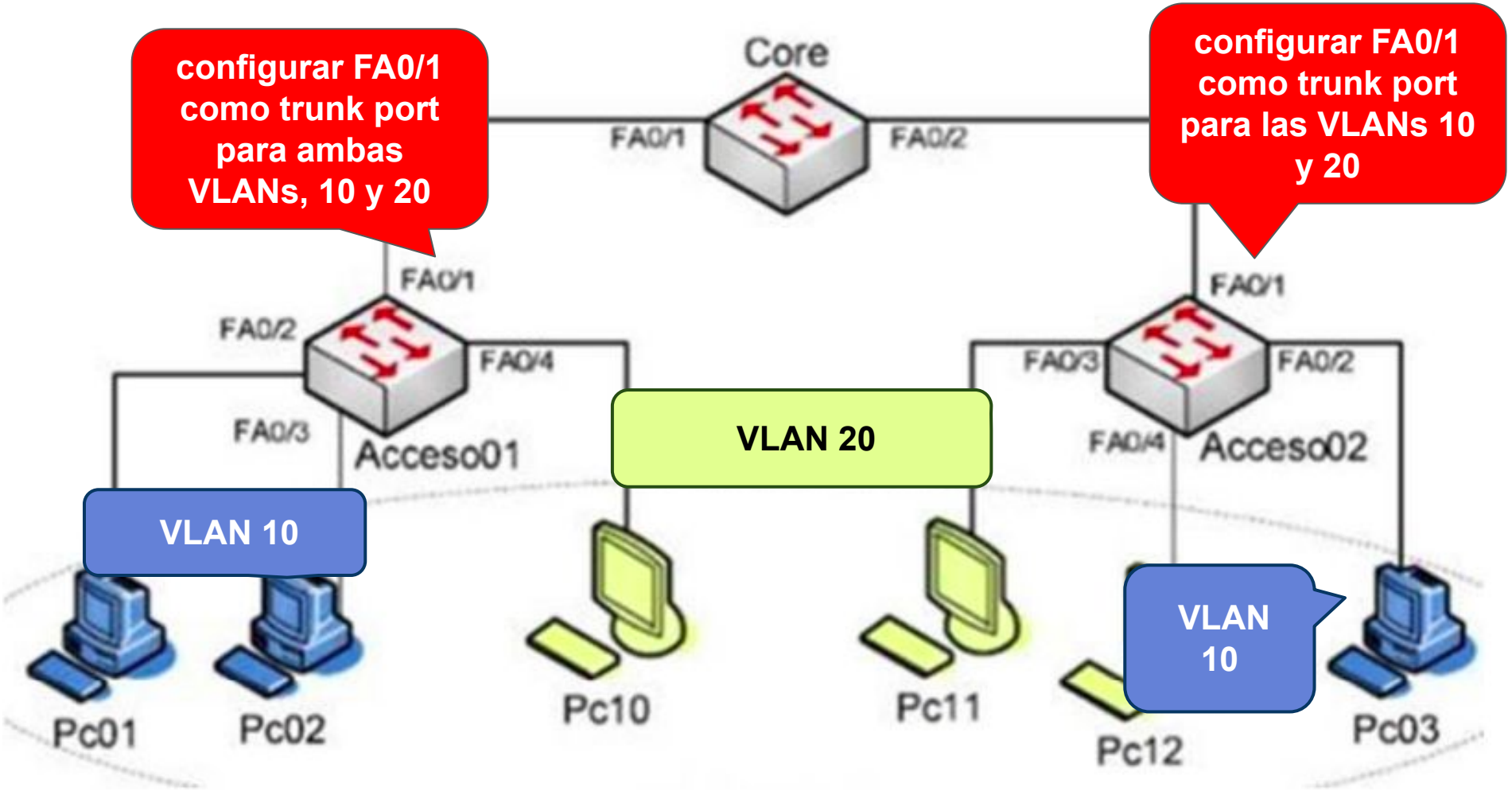
- **Access Port** - The frames received on the interface are assumed to not have a VLAN tag and are assigned to the specified VLAN. Access ports are used primarily for hosts and can only carry traffic for a single VLAN.

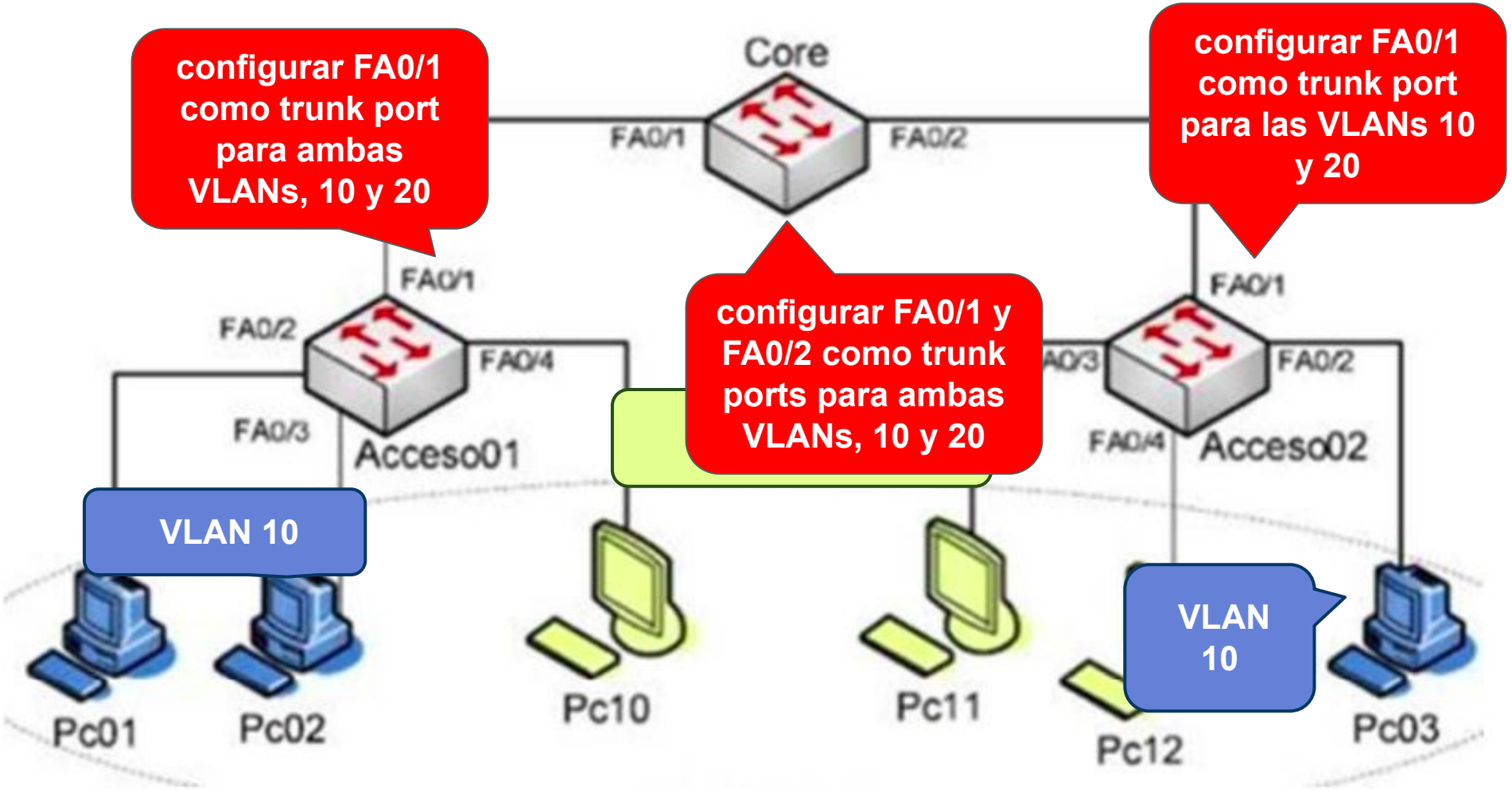
- **Trunk Port** - The frames received on the interface are assumed to have VLAN tags. Trunk ports are for links between switches or other network devices and are capable of carrying traffic for multiple VLANs.


Core
FA0/1
FA0/2
configurar FA0/2 como access port para la VLAN 10
FA0/1
FA0/2
FA0/4
FA0/3
Acceso01
VLAN 20
FA0/1
FA0/3
FA0/2
FA0/4
Acceso02
VLAN 10
VLAN 10
Pc01
Pc02
Pc10
Pc11
Pc12
Pc03


configurar FA0/2 y FA0/3 como access ports para la VLAN 10
configurar FA0/4 como access port para la VLAN 20
Core
configurar FA0/3 y FA0/4 como access ports para la VLAN 20
configurar FA0/2 como access port para la VLAN 10
FA0/1
FA0/2
FA0/4
FA0/3
Acceso01
VLAN 20
FA0/1
FA0/3
FA0/2
FA0/4
Acceso02
VLAN 10
VLAN 10
Pc01
Pc02
Pc10
Pc11
Pc12
Pc03


Core
FA0/1
FA0/2
configurar FA0/1 como trunk port para las VLANs 10 y 20
FA0/1
FA0/2
FA0/4
FA0/3
Acceso01
VLAN 20
FA0/1
FA0/3
FA0/2
FA0/4
Acceso02
VLAN 10
Pc01
Pc02
Pc10
Pc11
VLAN 10
Pc12
Pc03


configurar FA0/1 como trunk port para ambas VLANs, 10 y 20
Core
FA0/1
FA0/2
configurar FA0/1 como trunk port para las VLANs 10 y 20
FA0/1
FA0/2
FA0/4
FA0/3
Acceso01
VLAN 20
FA0/1
FA0/3
FA0/2
FA0/4
Acceso02
VLAN 10
VLAN 10
Pc01
Pc02
Pc10
Pc11
Pc12
Pc03


configurar FA0/1 como trunk port para ambas VLANs, 10 y 20
Core
FA0/1
FA0/2
configurar FA0/1 como trunk port para las VLANs 10 y 20
FA0/1
FA0/2
FA0/4
FA0/3
Acceso01
configurar FA0/1 y FA0/2 como trunk ports para ambas VLANs, 10 y 20
FA0/1
FA0/3
FA0/2
FA0/4
Acceso02
VLAN 10
VLAN 10
Pc01
Pc02
Pc10
Pc11
Pc12
Pc03

Apéndice. Cómo configurar VLANs en routers Cisco (link)



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Configure Port to VLAN Interface Settings on a Switch through the CLI

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Updated: February 15, 2020 **Document ID:** SMB5653

Bias-Free Language

Objective

A Virtual Local Area Network (VLAN) allows you to logically segment a Local Area Network (LAN) into different broadcast domains. In scenarios where sensitive data may be broadcast on a network, VLANs can be created to enhance security by designating a broadcast to a specific VLAN. Only users that belong to a VLAN are able to access and manipulate the data on that VLAN.

You can configure the ports and specify whether the port should be in access or trunk mode, and assign specific ports to VLANs. This article provides instructions on how to configure an interface VLAN as an access or trunk port on your switch through the Command Line Interface (CLI).

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Yes No

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